MCV4U1 Review Package - Solutions

**Part A: Multiple Choice**-Write the letter of your choice in the space provided.

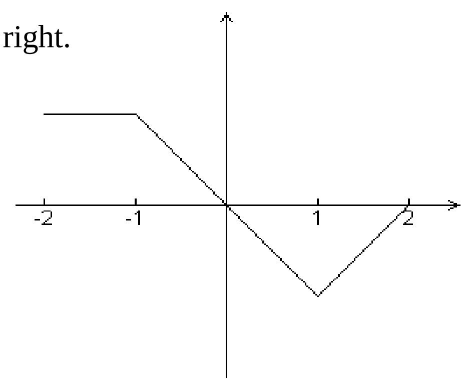
**D** ____ 1) Let *f* be the function with derivative given by $f'(x) = x^2 - \frac{2}{x}$. On which of the following intervals is *f* decreasing?

A) $(-\infty, -1]$ B) $(-\infty, 0)$ C) $[-1, 0)$ D) $(0, \sqrt[3]{2})$ E) $(\sqrt[3]{2}, \infty)$

**C** ____ 2) If the line tangent to the graph of the function *f* at the point $(1, 7)$ passes through the point $(-2, -2)$, then $f'(1)$ is

A) $-5$ B) 1 C) 3 D) 7 E) undefined

**B** ____ 3) The graph of $f'$, the derivative of the function *f*, is shown at right. Which of the following statements is true about *f* ?



Graph of $f'$

A) *f* is decreasing for $-1 < x < 1$.

B) *f* is increasing for $-2 < x < 0$.

C) *f* is increasing for $1 < x < 2$.

D) *f* has a local minimum at $x = 0$.

E) *f* is not differentiable at $x = -1$ and $x = 1$.

**C** ____ 4) Determine constants *a* and *b* such that the function $f(x) = x^3 + ax^2 + bx + c$ has a relative minimum at $x = 4$ and a point of inflection at $x = 1$.

A) $a = 1, b = 3$ B) $a = -3, b = 3$ C) $a = -3, b = -24$
D) $a = 3, b = 0$ E) $a = -6, b = 2$

**E** ____ 5) Which of the following are true for $h(x) = x^4 - 4x$?

I. *h* has a point of inflection at $x = 0$
II. *h* has an absolute minimum of -3
III. The second derivative test for local extrema is inconclusive

A) I and II B) I and III C) II and III D) I only E) II only

## Part B: Full Solutions

**1.** Find the exact value of $a$ such that the function $f(x)=\sqrt{x-2}-\frac{a}{x}$ has a point of inflection at $x$=3.

$$f(x)=(x-2)^{\frac{1}{2}}-ax^{-1}$$

$$f'(x)=\frac{1}{2}(x-2)^{-\frac{1}{2}}+ax^{-2}$$

$$f''(x)=\frac{-1}{4}(x-2)^{-\frac{3}{2}}-2ax^{-3}$$

$$f''(3)=0\rightarrow\frac{-1}{4}(3-2)^{-\frac{3}{2}}-2a(3)^{-3}=0$$

$$\frac{-1}{4}-\frac{2a}{27}=0$$

$$-27=8a\rightarrow\boxed{a=\frac{-27}{8}}$$

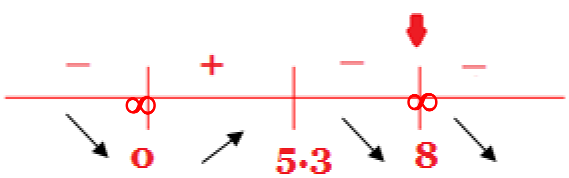
**2.** Sketch $f(x)=x^{\frac{2}{3}}(8-x)^{\frac{1}{3}}$. Find the coordinates of all relative extrema and inflection Points.

$$f'(x)=\frac{16-3x}{3x^{\frac{1}{3}}(8-x)^{\frac{2}{3}}}\qquad f''(x)=\frac{-128}{9x^{\frac{4}{3}}(8-x)^{\frac{5}{3}}}$$

$$f'(x)=0:16-3x=0$$

$$x=\frac{16}{3}\approx 5.3$$

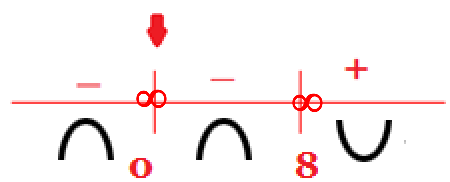
$$f'(x)\text{dne}\rightarrow x=0,x=8$$




local min: (0,0)

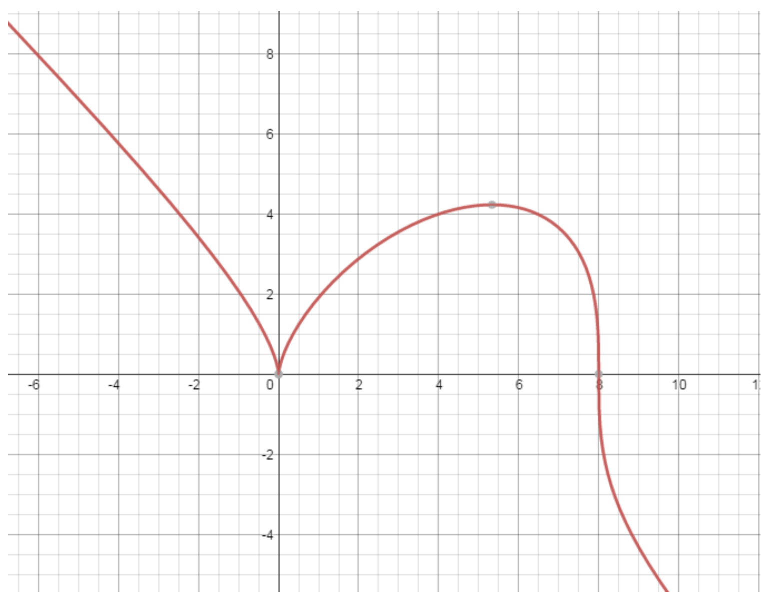
local max: (5.3,4.2)

$$f''(x)\text{dne}\rightarrow x=0,x=8$$




x-int: (0,0) &(8,0)

There is a cusp at x=0 and there is a vertical tangent at x=8



**3.** Perform a sketch for the following functions. Clearly indicate the results of each step.

**a)** $f(x)=\dfrac{x^3-4x}{3x^2+9x}$ , $f'(x)=\dfrac{x^2+6x+4}{3(x+3)^2}$ , $f''(x)=\dfrac{10}{3(x+3)^3}$

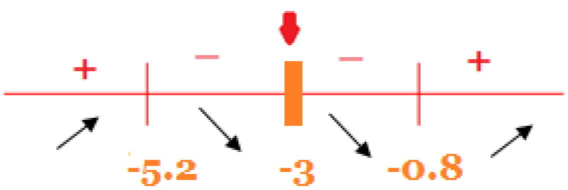
$$f(x)=\frac{\cancel{x}(x-2)(x+2)}{3\cancel{x}(x+3)}$$

$$f(x)=\frac{(x-2)(x+2)}{3(x+3)}, x\neq 0 \text{ (hole at } (0,\tfrac{-4}{9}))$$

**x-int :( 2,0)&(-2,0)**

**y-int: none**

$f'(x)=0: x=3\pm\sqrt{5}$




**local min:** $(-0.8,-0.5)$

**local max:** $(-5.2,-3.5)$

**V.A : x = -3**

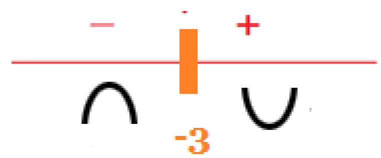
$\lim\limits_{x\to -3^-} f(x) = -\infty$ , $\lim\limits_{x\to -3^+} f(x)=\infty$

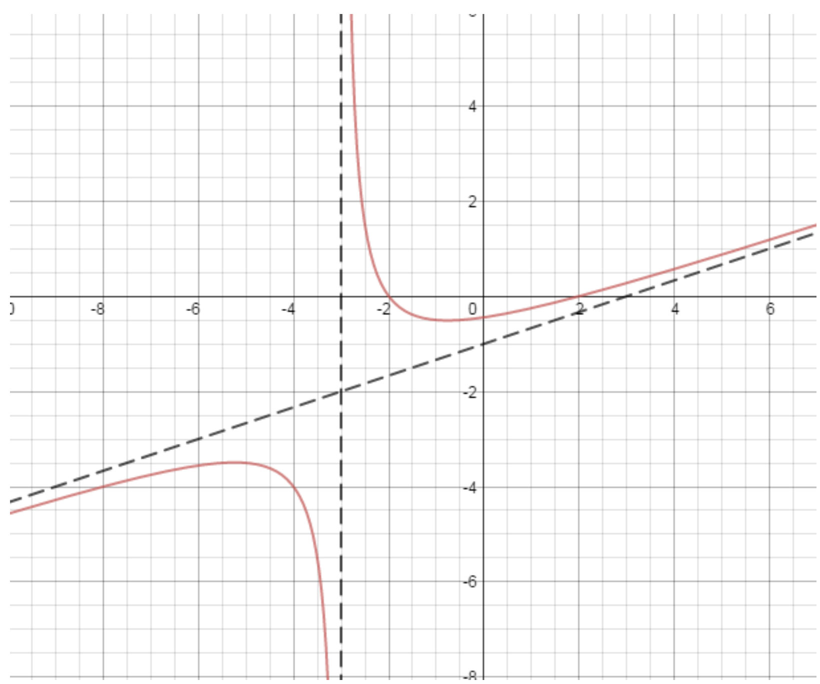
$$f(x)=(\tfrac{1}{3}x-1)+\frac{5}{3(x+3)}$$

**O.A:** $y=\frac{1}{3}x-1$

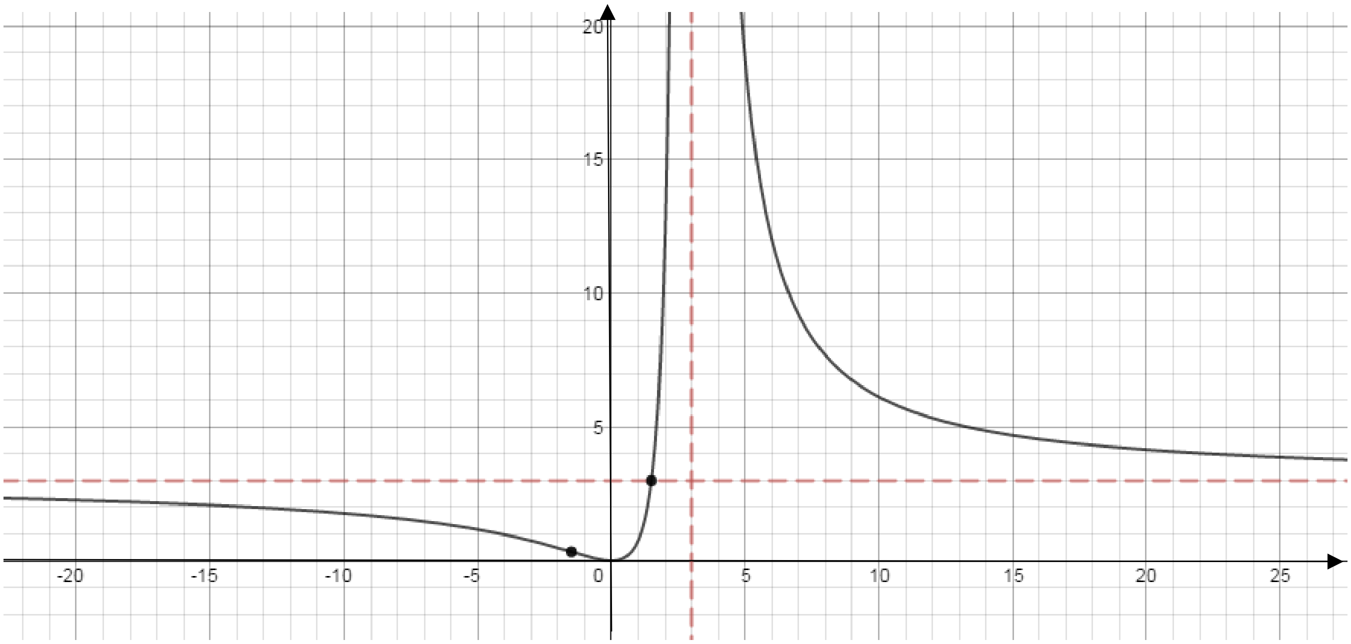
$\lim\limits_{x\to\infty}\left[f(x)-(\tfrac{1}{3}x-1)\right]=0$ **(above)**

$\lim\limits_{x\to-\infty}\left[f(x)-(\tfrac{1}{3}x-1)\right]=0$ **(below)**



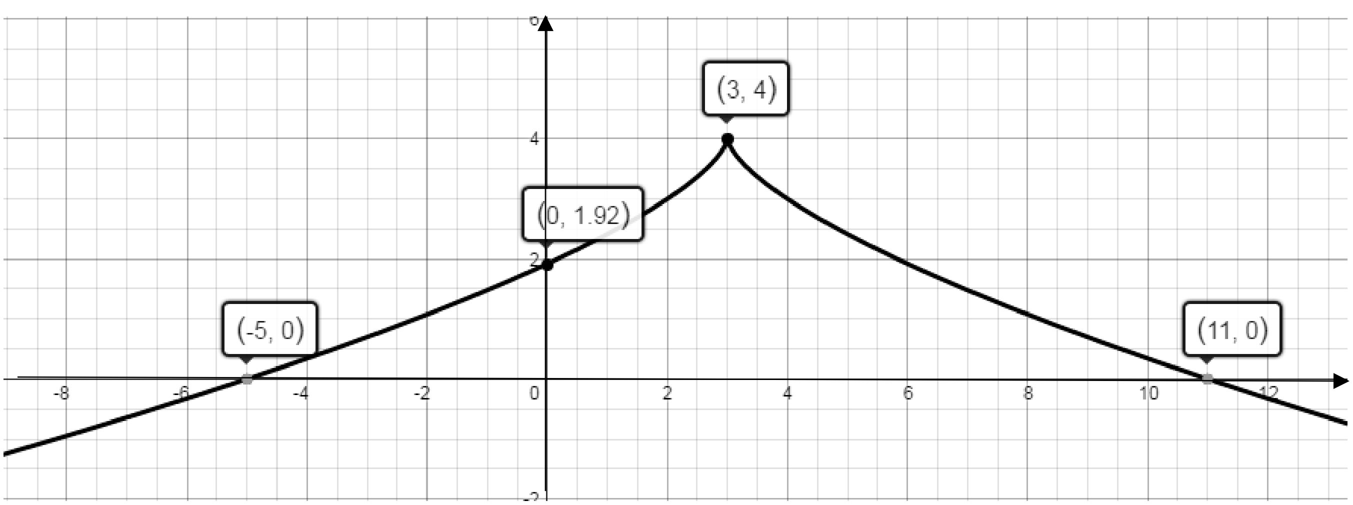
**b)** $f(x)=\dfrac{3x^2}{(x\text{-}3)^2}$ , $f'(x)=\dfrac{\text{-}18x}{(x\text{-}3)^3}$ , $f''(x)=\dfrac{18(2x+3)}{(x\text{-}3)^4}$




**c)** $f(x)=4-(x-3)^{\frac{2}{3}}$

$$f'(x)=-\frac{2}{3}(x-3)^{\frac{-1}{3}}$$

$$f''(x)=\frac{2}{9}(x-3)^{\frac{-4}{3}}$$



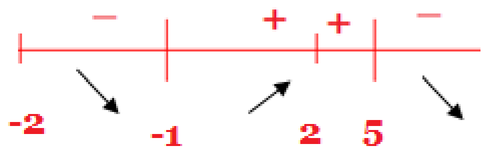
**4.** Consider the function $f(x) = -x^3 + 6x^2 + 15x + 2$ on the closed interval [-2, 2].

**a)** Find, and classify, all relative extrema.

**b)** Determine the absolute maximum of the function on the interval.

$$\mathbf{f'(x) = -3x^2 + 12x + 15}$$
$$\mathbf{0 = -3(x^2 - 4x - 5)}$$
$$\mathbf{0 = -3(x-5)(x+1)}$$
$$\mathbf{x = 5 \ , \boxed{x = -1}}$$

**N.I.D**




$$\mathbf{local\ min: (-1,-6)}$$
$$\mathbf{f(-2) = 4}$$
$$\mathbf{f(-1) = -6 \rightarrow abs.\ min}$$
$$\mathbf{f(2) = 48 \rightarrow abs.\ max}$$

**5.** Let $f(x) = x\left(4 + x^2 - \dfrac{x^4}{5}\right)$ .Find the interval(s) on which $f$ is increasing.

$$\mathbf{f(x) = 4x + x^3 - \frac{x^5}{5}}$$
$$\mathbf{f'(x) = 4 + 3x^2 - x^4}$$
$$\mathbf{0 = -(x^2 - 4)(x^2 + 1)}$$
$$\mathbf{0 = -(x-2)(x+2)(x^2 + 1)}$$
$$\mathbf{x = 2, -2}$$

$-\infty$ $-2$ $2$ $\infty$

| $x$ | | | |
|---|---|---|---|
| **Sign of $f'(x)$** | Negative | Positive | Negative |

$$\mathbf{Interval\ of\ increasing: (-2,2)}$$

**6.** Use the **SECOND DERIVATIVE TEST** to find the local extrema of the function $f(x)=3x^4+8x^3-6x^2-24x+3$ .

$f'(x)=12x^3+24x^2-12x-24$

$0=12(x^3+2x^2-x-2)$

$0=12(x-1)(x+1)(x+2)$

$x=1, x=-1, x=-2$

$f''(x)=12x^3+24x^2-12x-24$

$f''(x)=12(3x^2+4x-1)$

$f''(1)>0 \rightarrow$ **local min at** $x=1$

$f''(-1)<0 \rightarrow$ **local max at** $x=-1$

$f''(-2)>0 \rightarrow$ **local min at** $x=-2$

**7.** The function $f(x)=2kx^3+3x^2+px-3$ has a local minimum at x = -1 and a point of inflection at x =1 .Determine the values of *k* and *p*.

$f'(x)=6kx^2+6x+p$

$f''(x)=12kx+6$

$f'(-1)=0 \rightarrow 6k-6+p=0 \quad (1)$

$f''(1)=0 \rightarrow 12k+6=0 \quad (2)$

**From (2):** $\boxed{k=\frac{-1}{2}} \xrightarrow{\text{sun. into (1)}} -3-6+p=0$

$\boxed{p=9}$

**8.** A function is defined by $f(x)=ax^3+bx+c$.

**a)** Find the values of *a*, *b*, and *c* if $f(x)$ has a y-intercept at (0, 2) and a local maximum at (2, 6).

$f(x)=ax^3+bx+c$

$f'(x)=3ax^2+b$

$f(0)=2 \rightarrow \boxed{c=2}$

$f(2)=6 \rightarrow 8a+2b+c=6 \xrightarrow{\text{sub. c=2}} 8a+2b=4 \;\text{ or }\; 4a+b=2 \quad (*)$

$f'(2)=0 \rightarrow 12a+b=0 \longrightarrow b=-12a$

**sub.** $b=-12a$ **into (*):** $4a-12a=2$

$-8a=2$

$\boxed{a=\frac{-1}{4}} \;\&\; \boxed{b=3}$

**b)** Explain how you know there must be local minimum. **Using sign chart for *f'***

**9.** The slope of tangent to $g(x)=ax^3+bx^2+cx$ at its point of inflection $(1, 5)$ is 4. What are the values of a, b, and c.

$$\mathbf{g(x)=ax^3+bx^2+cx}$$
$$\mathbf{g'(x)=3ax^2+2bx+c}$$
$$\mathbf{g''(x)=6ax+2b}$$
$$\mathbf{g(1)=5\to a+b+c=5 \qquad (1)}$$
$$\mathbf{g'(1)=4\to 3a+2b+c=4 \qquad (2)}$$
$$\mathbf{g''(1)=0\to 6a+2b=0 \qquad (3)}$$
$$\mathbf{(2)-(1): 2a+b=-1 \qquad (4)}$$
$$\left.\begin{cases}\mathbf{3a+b=0}\\ \mathbf{2a+b=-1}\end{cases}\right\rangle \boxed{\mathbf{a=1}}\ \&\ \boxed{\mathbf{b=-3}}\ \&\ \boxed{\mathbf{c=7}}$$

**10.** Sketch the graph of a rational function that satisfies all of the following conditions:

$f''(x)<0$ *when* $x<-4$ $\quad$ $f''(x)>0$ *when* $x>-4$ $\quad$ $f'(x)<0$ *for all* $x$

$\lim\limits_{x\to-4^+} f(x)=+\infty$ $\quad$ $\lim\limits_{x\to-4^-} f(x)=-\infty$ $\quad$ $\lim\limits_{x\to\pm\infty} f(x)=-2$ $\quad$ $f(2)=0$